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PROBALISTIC VIEW OF LINEAR REGRESSION

$x \in \mathbb{R}^d$, $y \in \mathbb{R}$.

$$y|x \sim w^T x + \epsilon$$

$$\epsilon \rightarrow \text{noise} \sim N(0, \sigma^2)$$

$$\hat{w}_{ML} = (w^T x_i - y_i)^2 \,/\, (XX^T)^{-1} Xy$$

Goodness of $\hat{w}_{ML}$ in estimating w^*

$$E[\|w_{ML} - w\|^2] \Rightarrow \sigma^2 . \text{trace}\,(XX^T)^{-1}$$

Mean squared error

$$= \sigma^2 \left(\sum_{i=1}^{d} \frac{1}{\lambda_i} \right)$$

Bayesian

$$\hat{w}_{MAP} = \left(XX^T + \frac{1}{\gamma^2} I \right)^{-1} Xy$$

- MAP estimation for linear regression with a Gaussian Prior $N(0, \gamma^2 I)$ for w is equivalent to $\hat{w}_{new} = (XX^T + \lambda I)^{-1} Xy$

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Linear Regression

$$\hat{w}_{ML} = \arg\min_{w} \sum_{i=1}^{n} (w^T x_i - y_i)^2$$

Ridge Regression

$$\hat{w}_R = \arg\min_{w} \sum_{i=1}^{n} (w^T x_i - y_i)^2 + \underbrace{\lambda \|w\|^2}_{\text{regulariser}}$$

- A regulariser tends to reduce the weights of different features.
- Ridge regression will choose features that have the least weight.
- Ridge regression will apply more penalty if the length of vector w is large.

$$\min_{w \in \mathbb{R}^d} \sum_{i=1}^{d} (w^T x_i - y_i)^2 + \lambda \|w\|^2$$

$$\equiv$$

$$\min_{w \in \mathbb{R}^d} \sum_{i=1}^{d} (w^T x_i - y_i)^2$$

such that, $\|w\|^2 \leq \theta$ ← depends on λ

Closed form → $(XX^T + \lambda I)^{-1} Xy$

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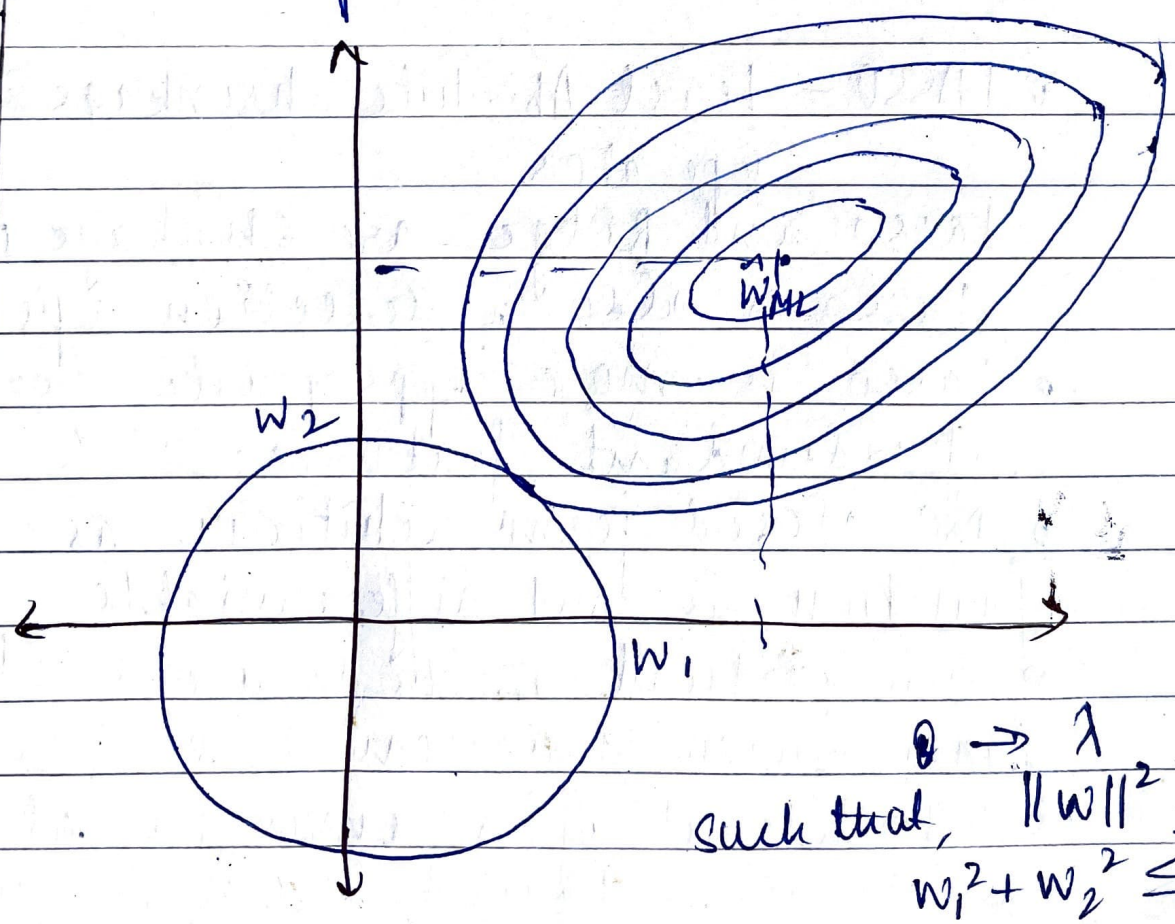
Linear Regression and Ridge Regression

- Ridge regression is a constrained optimisation problem where the constraint is applied on w.

- MSE produced by the vector w_R can be more than MSE produced by $\hat{w}_{ML}$ because it tries to minimise the sum of MSE and $\|w\|^2$.

- Ridge regression pushes weights of features towards zero but does not necessarily make them zero.



$\theta \rightarrow \lambda$

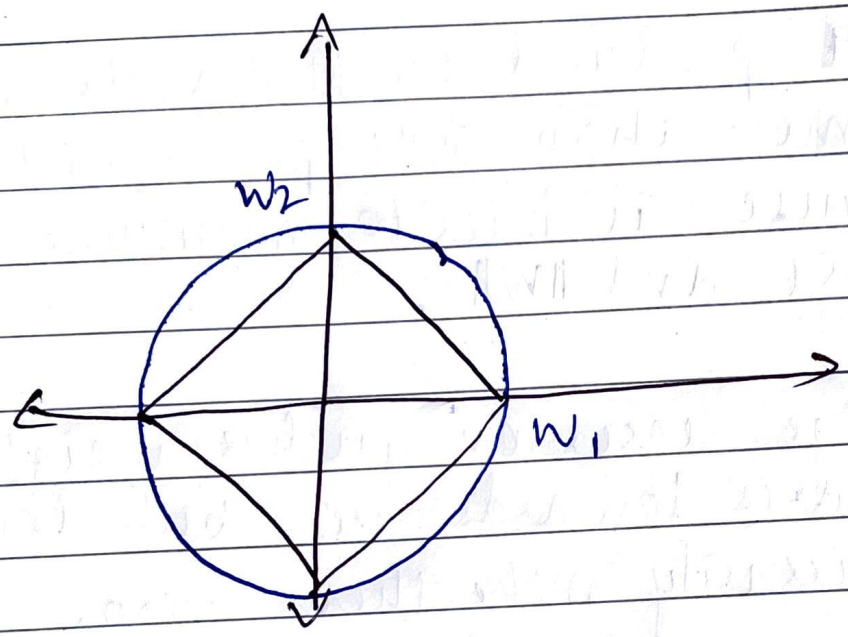
such that, $\|w\|^2 \leq \theta$

$w_1^2 + w_2^2 \leq \theta$

Lasso Regression

Loss fn: $\sum_{i=1}^{n} (w^T x_i - y_i)^2 + \lambda \|w\|_1$

[L1 norm is sum of absolute values]



- LASSO - Least Absolute Shrinkage & Selection Operator
- Lasso and Ridge are shrinkage operators; Lasso is also a selection operators
- Lasso is more appropriate ~~for~~ in case of redundant features.
- No closed form solution as error function is not differentiable
- Sub-gradient methods are used to optimize ~~Ridge~~ Lasso regression error function.
- Subgradient of a convex fn at a point always lower bounds the given function

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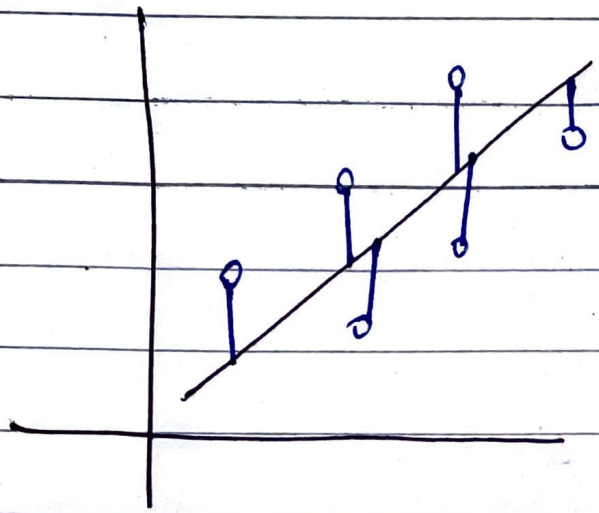
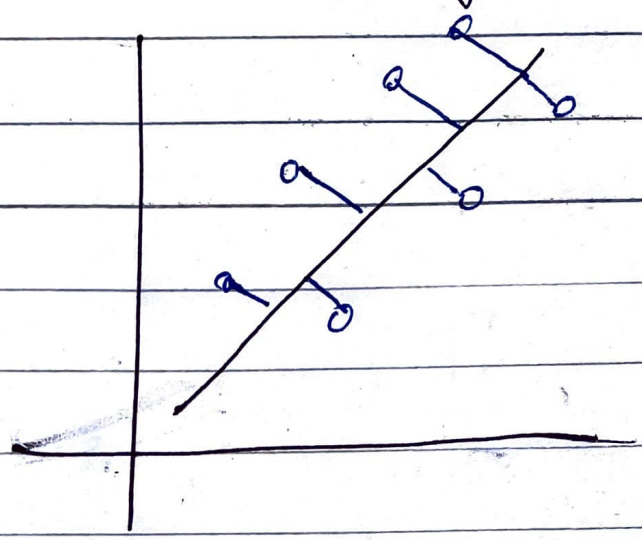
Error function → PCA vs Linear Regression



Linear Regression



PCA